

Blind Disturbance-Model Selection for Diffusion-Aided Semantic Image Reception

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Abstract—Explicit interference reconstruction is useful only when its additional modeling assumptions improve the desired image estimate. With unknown interference power, a joint diffusion sampler may instead introduce errors and unnecessary computation. We propose a blind disturbance-model selector for a frozen semantic image codec. Before diffusion inference, the receiver uses excess received energy and block-energy heterogeneity to choose either Gaussian denoising or joint signal–interference reconstruction. The rule is fitted on separate development images; only the selected branch is executed, and no neural weights are updated. We evaluate the rule on CelebA images with CIFAR-10 interference, using identical received signals for all receiver comparisons and a joint receiver with the same bypass as an additional control. On 256 independent image pairs with two random seeds and 18 channel conditions, the frozen rule attains 21.593 dB mean PSNR: +1.334 dB relative to calibrated joint sampling and +0.750 dB relative to the blind Gaussian receiver. It uses 52.2 total predictor evaluations per frame on average, compared with 80 for joint sampling. Separately trained DeepJSCC and MambaJSCC codecs provide external performance references under the same channel-use budget.

Index Terms—Semantic communications, interference cancellation, diffusion models, receiver adaptation, deep joint source–channel coding.

I. INTRODUCTION

Deep joint source–channel coding (JSCC) transmits an image by mapping its pixels directly to channel symbols. The encoder and decoder are trained together, enabling reconstruction quality to degrade gradually as the channel worsens [1]. SwinJSCC uses transformer features for image transmission [2], while MambaJSCC adopts a state-space architecture [3]. These learned codecs address the bandwidth and noise constraints of wireless image transmission. A deployed receiver, however, may also encounter co-channel interference whose strength and structure differ from the channel used in training.

Generative receivers can improve reconstruction without redesigning the transmitted representation. CDDM matches the channel disturbance to a diffusion noise level and denoises the received latent [4]. ICDM explicitly models the desired and interfering latents and recovers them jointly [5]. Receiver adaptation has also been studied through iterative source error correction [6], channel-conditioned diffusion reconstruction [7], and blind estimation of forward-model parameters [8]. These methods motivate using learned priors at reception, but leave a practical choice for a receiver equipped with both denoising and interference-reconstruction capabilities.

Explicit separation need not be preferable on every received frame. A Gaussian approximation ignores the interferer’s

structure, yet avoids estimating a second latent signal and its unknown amplitude. Joint reconstruction exploits that structure but must infer two signals from one mixture using imperfect priors and finite-step sampling. Errors in the inferred interferer can distort the desired signal even when the reconstructed mixture fits the observation. Always invoking the joint receiver can therefore increase both reconstruction error and computation. Conversely, always denoising the aggregate disturbance can discard structure that would help recover the image.

This paper studies when to use each receiver while keeping the codec and learned priors fixed. The choice must be made from the received symbols, without the clean image or true interference power. Evaluating both branches would incur both inference costs, and their true reconstruction errors are unavailable at reception. We instead use excess received energy and its variation across symbol blocks to select one branch before diffusion sampling. A threshold rule is fitted on separate development images, after which its parameters remain fixed. The selected latent estimate is passed to the common image decoder.

The contributions are as follows:

- We formulate blind disturbance-model selection between Gaussian denoising and joint signal–interference reconstruction for a frozen codec, requiring only $O(n)$ energy calculations before the chosen receiver runs.
- We fit energy-only and two-feature rules by image-group validation, with a fixed-receiver fallback when validation finds no selection gain.
- We evaluate the frozen rule on independent image pairs using identical received signals, a blind total-disturbance Gaussian baseline, and a joint receiver with the same direct-decoding gate; the comparison includes quality, inference cost, and harmful decisions.

The controls distinguish model selection from improved power estimation or simply bypassing separation. Separately trained DeepJSCC and MambaJSCC codecs provide external references, while public checkpoints test transfer without fine-tuning. The Gaussian and joint receivers share frozen priors and are receiver-only adaptations; this study does not reproduce the complete published CDDM or ICDM training pipelines.

Section II defines the transmission model and receiver decision. Section III presents the two candidates and selection rule, Section IV reports the experiments, and Section V concludes the paper.

II. SYSTEM MODEL AND PROBLEM FORMULATION

In this section, we introduce the image transmission model with co-channel interference and formulate the receiver-selection problem.

A. Image Transmission with Co-Channel Interference

We consider a desired image transmitter, a co-channel interferer, and a receiver that reconstructs the desired image. Let $\mathbf{u} \in [0, 1]^d$ denote a vectorized RGB image, where $d = 3H_{\text{img}}W_{\text{img}}$ for image height H_{img} and width W_{img} . The frozen source encoder f_s maps $\mathbf{u}$ to n complex channel symbols, and a separate encoder f_z maps an interfering image $\mathbf{v}$ to the same dimension:

$$\mathbf{x} = \mathcal{N}(f_s(\mathbf{u})), \quad \mathbf{z} = \mathcal{N}(f_z(\mathbf{v})), \quad \mathcal{N}(\mathbf{t}) = \frac{\sqrt{n} \mathbf{t}}{\|\mathbf{t}\|}. \quad (1)$$

Thus $\|\mathbf{x}\|^2 = \|\mathbf{z}\|^2 = n$, and the channel bandwidth ratio is n/d . The nonzero codewords are divided into B contiguous blocks, with L_b symbols in block b and $\sum_b L_b = n$. The additive channel is

$$\mathbf{y}_b = \mathbf{x}_b + a_b \mathbf{z}_b + \mathbf{w}_b, \quad \mathbf{w} \sim \mathcal{CN}(\mathbf{0}, N_0 \mathbf{I}), \quad a_b \geq 0, \quad (2)$$

Here a_b is the unknown interference amplitude in block b , and N_0 is the complex AWGN power. We assume unit desired-channel gain and known symbol alignment. The receiver knows N_0 , the normalization convention, and its block partition, but has no access to a_b , $\mathbf{z}$, or $\mathbf{u}$. For a prescribed nominal SINR γ , the simulator sets average nominal interference power to $10^{-\gamma/10} - N_0$. The instantaneous interference energy also depends on the realized block energies of $\mathbf{z}$; nominal SINR must therefore not be confused with measured per-frame SINR.

B. Receiver Decision

Let $\mathcal{R}_G$ and $\mathcal{R}_S$ denote Gaussian denoising and joint signal–interference reconstruction, respectively. A selector $\pi_\theta(\mathbf{y}) \in \{G, S\}$ chooses one receiver from the observation. The common frozen decoder g_s then reconstructs

$$\hat{\mathbf{u}}_\theta = g_s(\mathcal{N}(\mathcal{R}_{\pi_\theta(\mathbf{y})}(\mathbf{y}))). \quad (3)$$

The design objective is to maximize mean per-image PSNR over a development set by choosing θ , with all encoder, decoder and diffusion weights held fixed. Only the chosen branch is evaluated online. This separates the receiver decision from codec retraining and makes its additional inference cost explicit.

III. BLIND DISTURBANCE-MODEL SELECTION

In this section, we describe the two candidate receivers and develop a blind selection rule that chooses one receiver before diffusion inference.

A. Candidate Disturbance Models

The explicit model associates two learned priors with (2):

$$p(\mathbf{x}, \mathbf{z} | \mathbf{y}, \mathbf{a}) \propto p_s(\mathbf{x}) p_z(\mathbf{z}) \times \exp \left[-\frac{1}{N_0} \sum_b \|\mathbf{y}_b - \mathbf{x}_b - a_b \mathbf{z}_b\|^2 \right]. \quad (4)$$

The sampler approximates this posterior with finitely many updates. The mixture alone cannot identify both latents: the change $\mathbf{x}'_b = \mathbf{x}_b + a_b \delta_b$, $\mathbf{z}'_b = \mathbf{z}_b - \delta_b$ leaves the mixture unchanged. The Gaussian model instead uses

$$p_G(\mathbf{y} | \mathbf{x}; \nu) = \mathcal{CN}(\mathbf{y}; \mathbf{x}, \nu \mathbf{I}), \quad \nu \geq N_0. \quad (5)$$

It sacrifices interference structure to avoid explicit component estimation.

B. Candidate Receivers

Both receivers use the same frozen desired-signal prior and decoder. For Gaussian reception, estimate the disturbance from raw received symbols:

$$e_b = \frac{\|\mathbf{y}_b\|^2}{L_b}, \quad \bar{e} = \frac{1}{n} \sum_b L_b e_b, \quad (6)$$

$$\hat{P} = [\bar{e} - 1 - N_0]_+, \quad \hat{\nu} = N_0 + \hat{P}.$$

With unit source power and vanishing signal–interference cross expectation, unclipped excess energy estimates mean interference energy. Finite blocks, latent correlations, and clipping prevent treating it as an exact amplitude estimate. In particular, local desired-signal energy can resemble interference heterogeneity.

Let $r(\mathbf{y}) \in \mathbb{R}^{2n}$ stack real and imaginary components. Given the training diffusion schedule $\bar{\alpha}_t$, the Gaussian receiver G starts at the nearest noise level satisfying

$$\frac{1 - \bar{\alpha}_{t_0}}{\bar{\alpha}_{t_0}} \simeq \frac{\hat{\nu}}{2}, \quad \mathbf{v}_{t_0} = \sqrt{\bar{\alpha}_{t_0}} r(\mathbf{y}). \quad (7)$$

The factor 1/2 converts complex noise power to variance per real coordinate. For decreasing respaced indices, a noise predictor ϵ_s gives

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{v}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_s(\mathbf{v}_t, t)}{\sqrt{\bar{\alpha}_t}}, \quad (8)$$

$$\mathbf{v}_{t'} = \sqrt{\bar{\alpha}_{t'}} \hat{\mathbf{x}}_0 + \sqrt{1 - \bar{\alpha}_{t'}} \epsilon_s(\mathbf{v}_t, t). \quad (9)$$

At most 40 predictor evaluations are used, with fewer when the matched initial index is below 39.

The structured receiver S initializes $a_b^{(0)} = \sqrt{[e_b - 1 - N_0]_+}$ and performs joint diffusion inference with signal and interference predictors. During late corrector updates it obtains denoised, frame-normalized estimates $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{z}}$, then updates

$$a_b^+ = \left[\frac{\Re\{\tilde{\mathbf{z}}_b^H(\mathbf{y}_b - \tilde{\mathbf{x}}_b)\}}{\|\tilde{\mathbf{z}}_b\|^2} \right]_+. \quad (10)$$

An empty denominator retains the previous amplitude. This is a nonnegative least-squares update conditional on point

estimates, not a correction for their uncertainty. Calibration begins when the signal coefficient $\alpha(t) \geq 0.5$. Estimated block SINR updates the inherited guidance coefficients by clipped linear interpolation of the original AWGN table. We use the existing 40-step, order-three joint sampler: 40 signal and 40 interference evaluations. The initialization-only variant keeps amplitudes fixed and supplies an ablation of calibration.

C. Selection Before Generative Inference

The selector augments $\hat{P}$ with normalized block-energy variation

$$H = \frac{B^{-1} \sum_{b=1}^B (e_b - \bar{e})^2}{\max(\bar{e}^2, \varepsilon)}. \quad (11)$$

Here $\varepsilon > 0$ prevents division by zero, and all receiver blocks have equal length. H measures energy variation, not the interference distribution. For $\theta = (\tau_\ell, \tau_h, \eta)$, define

$$\pi_\theta(\mathbf{y}) = \begin{cases} \text{G}, & \hat{P} \leq \tau_\ell, H \leq \eta, \\ \text{G}, & \hat{P} \leq \tau_h, H > \eta, \\ \text{S}, & \text{otherwise.} \end{cases} \quad (12)$$

We also test the reversed assignment of G and S in (12); the Gaussian branch is not assumed to belong to the low-power region. The energy-only family sets $\tau_\ell = \tau_h$. Either family can reduce to a constant receiver; heterogeneity need not be useful.

We fit thresholds to mean per-image PSNR on development data. Eight folds keep all channel conditions belonging to one image together. Each fold is predicted using thresholds fitted on the other seven, then the family with the highest out-of-fold mean is refitted on all development images. Ties prefer lower component inference cost and then the simpler family. If neither family improves on the better fixed receiver in development validation, that fixed receiver is retained. No threshold is adjusted using confirmation or pressure results. At reception, (12) is computed first and only its chosen branch is executed. Selection adds $O(n)$ energy calculations and no diffusion evaluations.

For analysis, let $q_j(\mathbf{u}, \mathbf{y})$ be a candidate's PSNR. The unavailable per-frame oracle has quality $\max(q_G, q_S)$. The exact identity

$$\max(q_G, q_S) - q_{\pi_\theta} = |q_G - q_S| \mathbf{1}\{\pi_\theta \neq \arg \max_j q_j\} \quad (13)$$

shows that complementary receivers are necessary but not sufficient: mistakes on large-gap frames matter most. This is an accounting identity, not a generalization bound. Mean PSNR is the fitting objective; neither mean MSE, LPIPS nor harmful-tail probability is implicitly optimized.

IV. EXPERIMENTAL EVALUATION

In this section, we evaluate reconstruction quality and inference cost against fixed receivers and external codecs, followed by tests of harmful decisions and interference mismatch.

A. Experimental Setup

Desired images are 128×128 CelebA center crops [9]; 32×32 CIFAR-10 interference images are nearest-neighbor upsampled. Training uses 162,770 desired and 50,000 interference images. The frozen SwinJSCC codec has stage widths (128, 192, 256), depths (2, 2, 6), and an $8 \times 16 \times 16$ real latent: 1,024 complex symbols at CBR 1/48. Each DiT prior has 28 blocks, width 1,152, 16 heads and patch size two. The inherited training schedule uses 40/100 desired/interference-codec epochs and 50/20 desired/interference-prior epochs; selection adds no neural training.

DeepJSCC and MambaJSCC direct-codec references each train desired and interference codecs for 40 epochs at SNR 20 dB and CBR 1/48, using one seed and validation MSE for checkpoint selection. DeepJSCC uses a pinned third-party PyTorch reproduction; MambaJSCC uses official code. Public checkpoints are also evaluated without fine-tuning, with one encoder processing both image streams. Public MambaJSCC differs in training domain, channel and resolution; public DeepJSCC additionally uses CBR 0.16256. Their results are transfer diagnostics. The SwinJSCC-specific diffusion priors cannot be attached to these codecs without new prior training; complete published CDDM/ICDM systems are not reproduced here.

Development uses 64 validation pairs and seed 1700. Independent confirmation uses 256 test pairs, seeds 2700/2701, SNR 20 dB, and nominal SINRs $\{-7, -4, 0, 4, 7, 10\}$ dB. Sixteen 64-symbol blocks have stationary, alternating (0.25, 1.75), or burst (0, 2) relative interference powers. The 18 conditions give 9,216 frames per receiver. Confirmation follows policy freezing. File hashes exclude overlap with development, pressure tests and 200 previously evaluated pairs; independence is at image-pair, not identity, level. The power grid is $\{0, 0.1, 0.2, 0.35, 0.5, 0.75, 1, 1.5, 2, 3, 4, 8, 10^6\}$; the heterogeneity grid is $\{0.05, 0.1, 0.2, 0.4, 0.8, 1.6\}$, giving 26 energy-only and 2,028 two-feature rules across both assignment directions.

Controls include direct decoding, Gaussian reception using N_0 alone, blind Gaussian reception, initialization-only and calibrated joint sampling, and the selector. The same-gate control uses direct decoding for $\hat{P} \leq 0.15$ and calibrated joint sampling otherwise. Its outputs and component times are reconstructed from paired branches, excluding gate overhead. Development also tests coupled feedback and a Gaussian receiver given nominal disturbance power, a privileged diagnostic. All SwinJSCC variants share received tensors and sampler seeds; external codecs share image pairs, conditions and channel seeds but generate their own codewords.

Measurements use float32, batch one, TF32 disabled, four CPU threads and an RTX 4090. Synchronized receiver time includes selection, sampling and decoding; encoding, metric computation and loading are excluded. The number of function evaluations (NFE) counts calls to both priors. Quality metrics are per-image PSNR and MSE, LPIPS-VGG v0.1 [10], and four-scale MS-SSIM [11], averaged arithmetically after RGB

TABLE I
RECONSTRUCTION QUALITY AND RECEIVER COST ON THE
CONFIRMATION SET.

Receiver	PSNR	MSE $\times 10^3$	LPIPS	MS-SSIM	NFE	ms
SwinJSCC direct	18.67	29.14	0.440	0.697	0.0	8
DeepJSCC direct	17.39	24.71	0.575	0.684	0.0	1
MambaJSCC direct	16.92	44.13	0.463	0.613	0.0	10
Gaussian (N0)	18.83	28.70	0.438	0.700	18.0	225
Blind Gaussian	20.84	24.90	0.378	0.782	40.0	488
Init. joint	19.61	20.15	0.363	0.793	80.0	1110
Calibrated joint	20.26	18.49	0.362	0.802	80.0	1180
Joint + gate	20.45	18.43	0.361	0.801	65.9	974
Selector	21.59	16.45	0.336	0.832	52.2	701

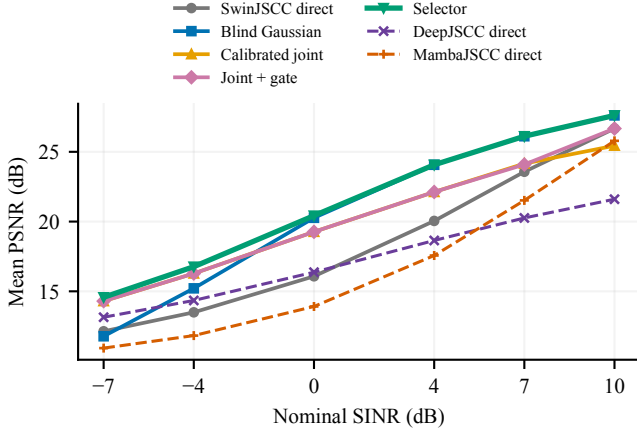


Fig. 1. Mean PSNR versus nominal SINR over the three interference profiles.

clipping to $[0, 1]$; four scales fit the image resolution. Paired 95% intervals use 5,000 bootstrap resamples of image pairs, retaining both seeds and all conditions. A harmful decision loses more than 0.5 dB against its comparator.

B. Frozen Rule and Receiver Comparison

Image-group validation yields 22.002 dB for energy-only selection and 22.239 dB for the two-feature family. The retained energy and heterogeneity rule uses $\tau_\ell = 8$, $\tau_h = 0.5$ and $\eta = 0.2$, with Gaussian reception below the applicable power threshold.

In the low-heterogeneity development region (810 frames), the largest observed excess-energy estimate is 5.241; the fitted threshold 8 therefore lies beyond observed support and acts as an always-Gaussian rule in that region. It is not an identified physical transition, and the family comparison does not establish that either feature is individually necessary. The earlier coupled-feedback receiver obtains 20.477 dB, versus 20.605 dB for calibrated joint sampling and 20.835 dB for its same-gate control. The privileged nominal-power Gaussian diagnostic obtains 21.591 dB. These development comparisons motivate the receiver decision but are not independent confirmation. Table I and Fig. 1 report the independent experiment. The selector changes mean PSNR by +1.334 dB relative to calibrated joint sampling (95% interval $[1.265, 1.404]$), +0.750 dB relative to blind Gaussian denoising (95% interval $[0.698,$

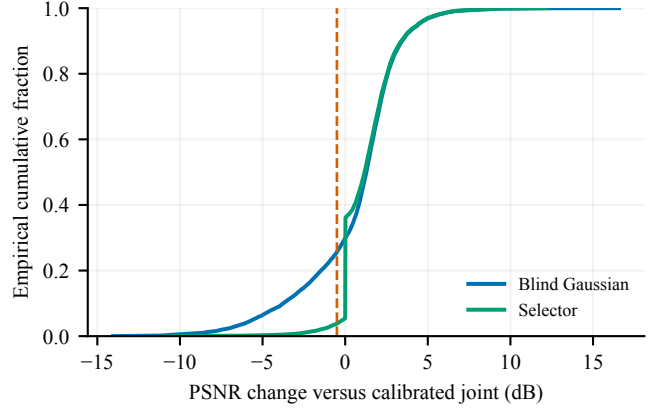


Fig. 2. Empirical CDF of per-frame PSNR change relative to calibrated joint sampling.

0.806]), and +1.140 dB relative to the same-gate control (95% interval $[1.075, 1.205]$). Both fixed-receiver comparisons have positive paired intervals, supporting a mean-PSNR benefit within the matched confirmation population. Estimating total disturbance changes the Gaussian receiver from 18.829 to 20.843 dB, so the thermal-noise-only port is insufficient as the sole Gaussian baseline.

The fixed receivers are complementary across interference profiles. For stationary interference, Gaussian denoising reaches 21.310 dB against 19.282 dB for joint sampling. Under burst interference, joint sampling reaches 21.317 dB against 20.481 dB for Gaussian denoising. The selector reaches 21.317, 21.285, and 22.177 dB for stationary, alternating, and burst interference, respectively. Thus its gain is consistent with using different receivers for different observations; the energy statistic alone does not establish a physical explanation of the interference.

Separately matched direct-codec anchors attain 17.394 dB for DeepJSCC and 16.923 dB for MambaJSCC; these use independently trained latent spaces and provide external codec references. One training seed and a fixed 40-epoch budget do not establish convergence or a general architecture ranking. Without fine-tuning, public DeepJSCC (ImageNet/SNR 19), MambaJSCC (CLIC2021/AWGN 10), and MambaJSCC (DIV2K/Rayleigh) weights yield 13.755, 17.020, and 16.650 dB, respectively. Their protocol mismatches preclude a controlled claim about retraining benefit. In Fig. 1, solid curves use the frozen SwinJSCC latent space, whereas dashed curves use the separately retrained codecs.

C. Inference Cost and Harmful Decisions

The selector chooses Gaussian reception on 69.5% of confirmation frames. Its average cost is 52.2 predictor evaluations and 701 ms, versus 80 evaluations and 1180 ms for joint sampling. Against that receiver, 3.79% of frames lose more than 0.5 dB, and the fifth percentile of the paired change is -0.127 dB (Fig. 2). The unavailable per-frame oracle reaches 21.760 dB; selection leaves 0.167 dB mean oracle regret.

TABLE II
MEAN PSNR UNDER BLOCK MISMATCH AND NO INTERFERENCE.

Profile	Direct	Gaussian	Joint	Select
Shifted blocks	20.83	22.99	21.09	23.00
Random lengths	20.64	23.42	22.74	23.98
No interference	31.37	31.37	27.73	31.37

Mean MSE changes from 0.01849 to 0.01645, LPIPS from 0.3618 to 0.3362, and MS-SSIM from 0.8015 to 0.8318. These secondary metrics and the harmful tail must accompany the average PSNR comparison. The CDF in Fig. 2 uses received frames; its vertical line marks the -0.5 dB harmful-loss threshold.

D. Block Mismatch and No Interference

Pressure tests use 64 further test pairs and seed 3700, with nominal SINR $-4, 4, 10$ dB. One profile shifts burst boundaries by half a receiver block; another draws segment lengths uniformly from 16 to 128 symbols and normalizes the binary on/off amplitude pattern. A separate SNR-20 condition has no interference. The receiver keeps its original block partition and frozen rule. These families are reported separately from matched confirmation. Table II shows a $+0.010$ dB selector change against Gaussian reception for shifted blocks and $+0.563$ dB for random lengths. With no interference, selection matches Gaussian reception and avoids the joint receiver's loss. These are diagnostic populations; no rule is refitted or result pooled with matched confirmation. For a visual check of confirmation, Fig. 3 shows all six prespecified examples: image pairs 512 and 527, each under conditions 0, 8 and 15; none was selected by quality.

V. CONCLUSION

In this paper, we proposed a blind selector that uses received energy and block-energy heterogeneity to choose between Gaussian denoising and joint signal–interference reconstruction with frozen neural weights. Independent confirmation shows PSNR gains of 0.750 and 1.334 dB over the respective receivers, with 34.7% fewer predictor evaluations than joint sampling. These gains are population-specific: harmful decisions remain, and block mismatch nearly removes the advantage over Gaussian denoising. Future work will address fading, synchronization errors, and task-oriented evaluation for vehicular links.

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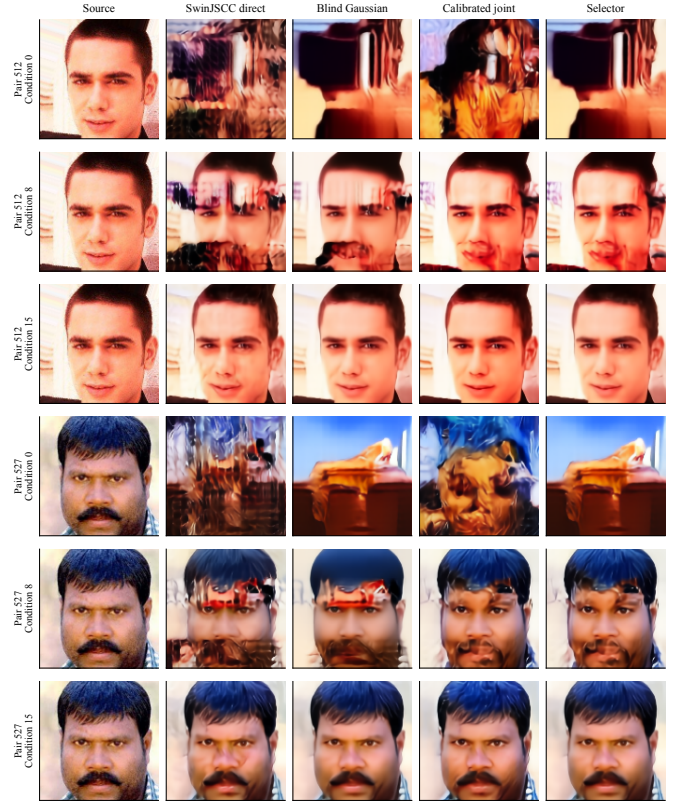


Fig. 3. Reconstructed images for six prespecified confirmation examples.

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